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First Late Cambrian Paleomagnetic Results From the Oulongbuluke Terrane, Northern Tibetan Plateau: Implications for the Paleogeography of the Proto-Tethys Ocean

Yong Cao[1], Junling Pei[2], Zhiming Sun[1], Haibing Li[3], Wei Xu[4], Xiaozhou Ye[1], and Lei Zhang[3]

[1]Key Laboratory of Paleomagnetism and Tectonic Reconstruction of Ministry of Natural Resource, Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing, China, [2]State Key Laboratory of Nuclear Resources and Environment, East China University of Technology, Nanchang, China, [3]Key Laboratory of Continental Dynamics of Ministry of Natural Resources, Institute of Geology, Chinese Academy of Geological Sciences, Beijing, China, [4]Xi'an Center, China Geological Survey (Geoscience Innovation Center of Northwest China), Xi'an, China

Key Points:

- We report the first Late Cambrian paleomagnetic results for the Oulongbuluke terrane (OT)
- The OT was located at ~20.6°S during the Late Cambrian
- The terranes in the northern Tibetan Plateau may have been located northwest of the Indian plate during the Late Cambrian

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

J. Pei,
jlpei@qq.com

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Author Contributions:

Conceptualization: Yong Cao, Junling Pei, Zhiming Sun, Haibing Li
Data curation: Yong Cao, Junling Pei, Zhiming Sun, Wei Xu, Xiaozhou Ye, Lei Zhang
Funding acquisition: Junling Pei, Zhiming Sun
Investigation: Yong Cao, Junling Pei, Zhiming Sun, Wei Xu, Xiaozhou Ye, Lei Zhang
Methodology: Yong Cao, Junling Pei, Zhiming Sun
Resources: Junling Pei
Writing – original draft: Yong Cao
Writing – review & editing: Junling Pei, Zhiming Sun, Haibing Li

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Abstract

The Early Paleozoic paleolatitudinal position of the terranes in the northern Tibetan Plateau is the key to unraveling the evolution of the Proto-Tethys Ocean. We present the first Late Cambrian paleomagnetic results from the Oulongbuluke terrane (OT), in the northern Tibetan Plateau. The mean paleomagnetic direction for 16 sites is $Ds = 196.2°$, $Is = -36.9°$ with $\kappa s = 31.5$, $\alpha_{95} = 6.7°$, corresponding to a paleopole at 68.2°S/51.9°E with dp/dm = 4.6°/7.8°. Rock magnetic and petrologic analyses demonstrate the primary origin of the magnetic mineralogy. This paleomagnetic result defines the Late Cambrian paleolatitude of the OT as 20.6 ± 4.6°S (reference point: 37.2°N/96.6°E). Combined paleomagnetic and geological evidence suggests that the terranes in the northern Tibetan Plateau were located to the northwest of the Indian plate of Gondwana during the Late Cambrian.

Plain Language Summary

The Proto-Tethys Ocean underwent a complex process of collision and amalgamation of the terranes in the northern Tibetan Plateau during the Early Paleozoic. Unraveling the Early Paleozoic paleogeography of these terranes is important for understanding the evolution of the Proto-Tethys Ocean and is critical for reconstructing the formation of East Asia. However, quantitative constraints on the Early Paleozoic paleogeographic position of the terranes in the northern Tibetan Plateau are limited. Paleomagnetism is the only effective approach for directly constraining the paleolatitude of terranes. We conducted a paleomagnetic study of the Late Cambrian strata in the Oulongbuluke terrane (OT), northern Tibetan Plateau. The results define the paleolatitude of 20.6 ± 4.6 °S for the OT during the Late Cambrian. Based on paleomagnetic data and geological evidence, we also present a tentative paleogeographic reconstruction of the terranes in the northern Tibetan Plateau and the Proto-Tethys Ocean in a global setting during the Late Cambrian.

1. Introduction

The Proto-Tethys Ocean was an ancient ocean that extended from the Tarim-North China Block in the north to the North Qiangtang-Songpan Ganze terrane in the south (Dong et al., 2021; Li et al., 2018; Song et al., 2018). Knowledge of the evolution of the Proto-Tethys Ocean is important for understanding the formation of the East Asian continent. The Proto-Tethys Ocean experienced a complex process of collision and amalgamation of the micro-continental terranes in the northern Tibetan Plateau, such as the Qaidam, Qilian, Oulongbuluke, and Kunlun terranes, in space and time (Dong et al., 2021; Li et al., 2018; Wu et al., 2020). Unraveling the paleogeography of these micro-continental terranes is important for determining the evolution of the Proto-Tethys Ocean and is critical for reconstructing the formation of East Asia. However, the Early Paleozoic paleogeographic positions and affinities of these terranes remain ambiguous.

The comparable characteristics of the Ediacaran-Cambrian lithostratigraphic sequence and paleontological assemblages suggest that the Oulongbuluke terrane (OT) may have been located close to the North China block (NCB) during the late Ediacaran to the Cambrian (Pang et al., 2021; Sun Z et al., 2024; Xiang et al., 1981; Yu et al., 2017). The similarities between the Early Paleozoic marine red beds in the OT and the NCB imply that the

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OT may be part of the NCB (Li, Lin, et al., 2019). However, the Cambrian trilobite assemblages of the Qilian Mountain are closely related to those of the South China block (SCB) (Lin et al., 2015; Zhou et al., 1996; Zhu et al., 2022). The zircon U-Pb age distributions of the Cambro-Ordovician sediments from the Qilian tectonic belts and the OT show strong similarities with coeval sediments in South China, Qiangtang, and Himalaya, implying the paleogeographic connection of the terranes in the northern Tibetan Plateau with the SCB or the Indian plate (Li M et al., 2019b; Li XY et al., 2022; Song et al., 2013; Sun et al., 2020; Wang et al., 2023; Yu et al., 2021). Furthermore, a synthesis of magmatic studies suggested that the terranes in the northern Tibetan Plateau had paleogeographic connections with the SCB during the Early Paleozoic (Peng et al., 2019).

The above overview shows that both the Early Paleozoic paleogeographic positions and affinities of the terranes in the northern Tibetan Plateau are controversial; furthermore, quantitative constraints on the paleogeographic position of the terranes are lacking. Paleomagnetism is the only effective approach for directly constraining the paleolatitudinal position of terranes, and the OT is the only terrane in the northern Tibetan Plateau for which the Early Paleozoic strata are relatively continuous and the sedimentary age is well constrained (Sun et al., 2020). Hence, this offers an opportunity to obtain paleomagnetic constraints on the paleogeography of the Proto-Tethys Ocean. However, until now, only two Early Paleozoic paleomagnetic results are available for the OT. The paleomagnetic results reported by Yang et al. (1992) indicate that the Early Ordovician paleolatitude for the OT was ∼13°S; however, their Early Ordovician paleomagnetic results are derived from only 10 samples and they lack field tests. A new Early Ordovician paleomagnetic result was recently reported by Cao et al. (2021). This result also lacked field tests, but the characteristic remanent magnetizations are all reverse polarities and are consistent with the Moyero Reversed Polarity Superchron, which indicates the primary origin of the characteristic remanence. This result indicates that the OT was located at a low latitude near the equator during the Early Ordovician (Cao et al., 2021).

Here we report the first Late Cambrian paleomagnetic results for the OT. These paleomagnetic results, combined with geological evidence, provide reliable constraints on the paleogeographic affinity of the terranes in the northern Tibetan Plateau, and the paleogeography of the Proto-Tethys Ocean during the Late Cambrian.

2. Materials and Methods

The OT is located in the northern Tibetan Plateau in northwest China (Figure 1a). It is a long and narrow belt in the area of Quanji Shan-Delingha-Wulan. The OT is separated from the Qaidam terrane in the south by the high-pressure ultrahigh-pressure metamorphic zone along the northern margin of the Qaidam basin, and in the north by faults in the newly defined metamorphic-magmatic unit of the Early Paleozoic Arc (Figure 1b). The study area is located in the Huaitoutala area in the southeastern region of Oulongbuluke Mountain, where the Paleozoic and Cenozoic strata are exposed. The Paleozoic mainly comprises the Cambrian Oulongbuluke Formation, the Lower Ordovician Duquanshan Formation, the Middle Ordovician Shihuigou Formation, the Lower Carboniferous Chengqianggou Formation and Huaitoutala Formation, and the Upper Carboniferous Zhabusagaxiu Formation. The Cenozoic mainly comprises the Paleocene Ganchigou Formation, the Neocene Shangyoushashan Formation, the Shizigou Formation, and the Quaternary (Figure 1c).

The samples in this study were collected from the Upper Oulongbuluke Formation (37.2°N/96.6°E). The Upper Oulongbuluke Formation, consisting of dark gray and purple-red limestone (Figure S1 in Supporting Information S1), is conformably in contact with the underlying Middle Oulongbuluke Formation and the overlying Ordovician Duquanshan Formation. Trilobite fossils including *Ulania quaitrata*, *Parakotuia* sp., *Anomocarella* sp., *Blackwelderia* sp., *Chuangia* sp., and *Parakotaia truncata* were found in the Upper Oulongbuluke Formation and indicate a Late Cambrian Furongian age (Chen et al., 2013; Sun et al., 2020). In this study, ∼150 paleomagnetic core samples were collected from 20 sites in the dark gray limestone of the Upper Oulongbuluke Formation. The cores were drilled in the field using a gasoline-powered drill and oriented with a magnetic compass. The methods employed for petrographic, rock magnetic, and paleomagnetic analyses are described in the Text S1 in Supporting Information S1.

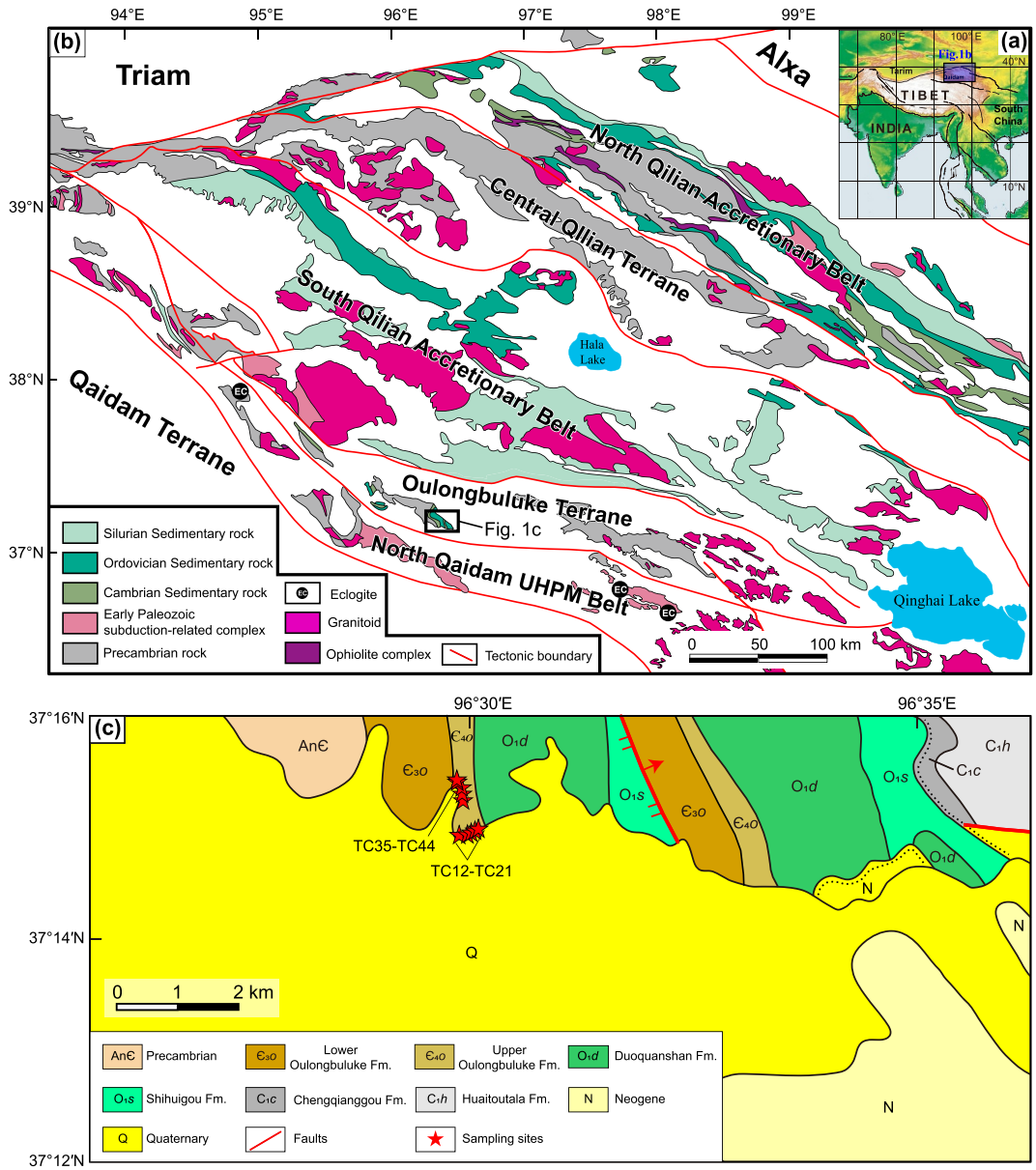


Figure 1. Geologic maps of the northern Tibetan Plateau and its adjacent area. (a) Location of the study area (the rectangle) in relation to the Tibetan Plateau and its adjacent areas, (b) Geologic map of the Oulongbuluke terrane and its adjacent area (modified from Sun et al. (2020)), (c) Simplified geologic map of the study area (modified from the Tuosuhu regional geological map).

3. Results

3.1. Petrography and Rock Magnetism

Scanning electron microscope (SEM) observations of representative samples of the Late Cambrian limestone from the OT show that the magnetic minerals are predominantly euhedral to subhedral in shape, with a grain size of ~10 μm (Figure S2 in Supporting Information S1). Energy-dispersive X-ray spectrometry (EDS) analyses show that the elemental composition of the magnetic minerals is mainly Fe, O, and Ti. Combined with the rock magnetic results, the SEM observations indicate that the main magnetic carrier in these limestone samples is Ti-bearing magnetite. The magnetic grains are irregular in shape, have well-defined boundaries with the groundmass, and lack oxidized rims (Figure S2 in Supporting Information S1); these characteristics indicate that they are

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unaffected by chemical alteration after diagenesis. The microstructural features of the magnetic minerals reveal that they are of detrital origin and are not autogenetic minerals.

The hard, intermediate, and soft magnetic components of the samples have an unblocking temperature of ~580°C (Figure S3 in Supporting Information S1), indicating that the primary magnetic carrier for these limestones is magnetite. The soft and hard components of sample TC13-2 and the soft, medium, and hard components of sample TC15-4 exhibited an unblocking temperature of 120°C (Figure S3 in Supporting Information S1), indicating the existence of goethite. The saturation magnetization (M_s) for the samples ranges from 2.2×10^{-5} Am^2/kg to 26.5×10^{-5} Am^2/kg, and the saturation remanence (M_r) ranges from 0.3×10^{-5} Am^2/kg to 13.2×10^{-5} Am^2/kg. M_r/M_s ratios range from 0.11 to 0.33. The values of the coercivity (B_c) and remanent coercivity (B_{cr}) range from 9.1 to 26.2 mT and from 23.8 to 75.3 mT, respectively. B_{cr}/B_c ratios range from 2.3 to 3.05. The low B_c and B_{cr} values indicate the dominance of low-coercivity magnetic minerals. The hysteresis loops are wasp-waisted (Figure S4 in Supporting Information S1), indicating the coexistence of low-coercivity and high-coercivity magnetic minerals in the Late Cambrian limestone in the OT (Roberts et al., 1995). This result is consistent with the results of the thermal demagnetization of a three-axis isothermal remanent magnetization. In the Day plot, the hysteresis data plot within the PSD region (Day et al., 1977), near the single-domain (SD) + multidomain (MD) mixing curve (Dunlop, 2002).

3.2. Paleomagnetism

The thermal demagnetization results for the Late Cambrian limestone from the OT show that most of the samples have both low-temperature components (LTC) and high-temperature components (HTC) (Figure 2). The LTC was isolated below ~240°C. The mean direction of the LTC obtained from the 20 sites is $Dg = 14.4°$, $Ig = 55.7°$, $\kappa g = 45.2$, $\alpha_{95} = 4.9°$ (in geographic coordinates), and $Ds = 38.0°$, $Is = 52.6°$, $\kappa s = 31.5$, $\alpha_{95} = 5.9°$ (in stratigraphic coordinates). The mean direction of the LTC in geographic coordinates overlaps with the direction of the present-day geomagnetic field (D/I = 359.3°/57.4°) at the sampling site (37.2°N/96.6°E) (Figure 3a), indicating that the LTC is likely a viscous remanent magnetization acquired in the present-day geomagnetic field.

For most of the limestone samples, the HTC shows linear relationships in the demagnetization curves and trends toward the origin (Figures 2a–2u). These HTC were isolated between 300°C and 580°C using principal component analysis (PCA, Kirschvink, 1980). The result of Elongation/Inclination (E/I) correction (Tauxe et al., 2008) of the HTC isolated by PCA indicates that the original inclination of the HTC falls within the range of the corrected confidence interval (Figure S4 in Supporting Information S1). This result indicates that no significant inclination shallowing has occurred in the limestone in the studied section. The HTC of a few samples (~18 samples) did not decay toward the origin but instead were distributed along a great circle in the equal-area projection plot (Figures 2v–2x), and these samples were analyzed using remagnetization circle analysis (Halls, 1978). The HTC direction were isolated for a total of 18 paleomagnetic sampling sites except for sites TC38 and TC39; however, the α_{95} values of sites TC21 and TC43 are >15° (Table 1), and therefore these two sites were excluded from the Fisher statistics. The mean direction of the HTC for the remaining 16 sites is $Dg = 184.0°$, $Ig = -35.5°$, $\kappa g = 31.5$, and $\alpha_{95} = 6.7°$ (in geographic coordinates), and $Ds = 196.2°$, $Is = -36.9°$, $\kappa s = 31.5$, $\alpha_{95} = 6.7°$ (in stratigraphic coordinates) (Table 1, Figure 3b). The corresponding paleopole lies at 68.2°S/51.9°E with dp/dm = 4.6°/7.8°.

4. Discussion

4.1. Late Cambrian Paleolatitude of the Oulongbuluke Terrane

Due to the monoclinal bedding attitude in the studied section, a fold test could not be conducted on the paleomagnetic results. However, both normal and reverse polarities were obtained from the HTC direction and the site mean direction passed the reversal test ($r = 11.9° < rc = 13.1°$) (McFadden & McElhinny, 1990). Microstructural analyses indicate that the magnetic minerals are clastic in origin and the rocks are not affected by alteration after diagenesis. The Day plot has been used to distinguish non-remagnetized and remagnetized carbonate rocks (Jackson & Swanson-Hysell, 2012; Roberts et al., 2013). The hysteresis parameters of carbonate rocks carrying a primary remanent magnetization mostly plotted close to the SD + MD mixing curve, while the hysteresis parameters of remagnetized carbonate rocks mostly plot near the SP + SD mixing curve (Jackson & Swanson-Hysell, 2012; Roberts et al., 2013). The hysteresis parameters of the limestone in this study plot close to the SD + MD mixing curve in the Day plot (Figure S3 in Supporting Information S1). This is consistent with the

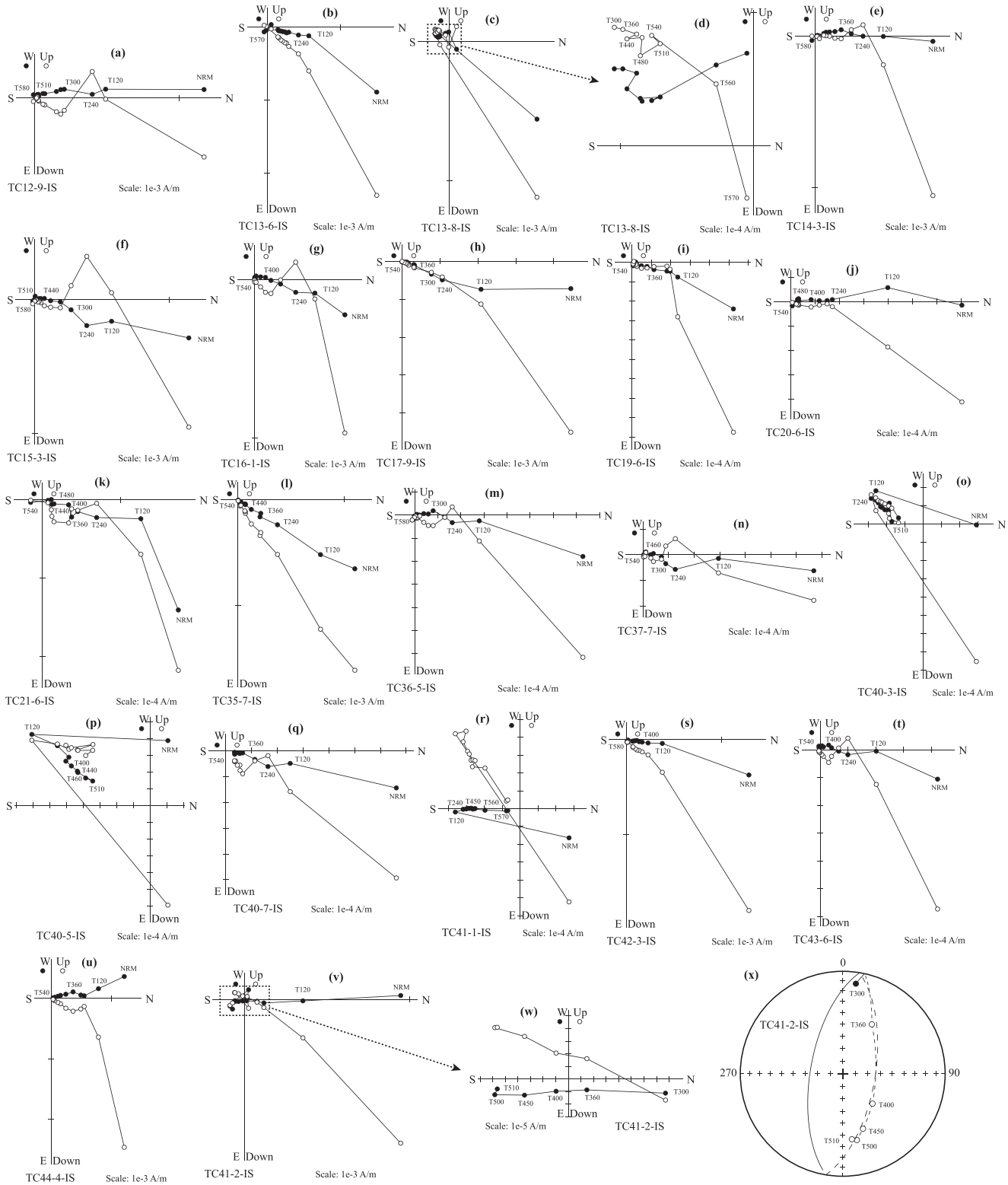


Figure 2. Stepwise thermal demagnetization diagrams of the Late Cambrian limestone from the Oulongbuluke terrane. (a–w) Zijderveld diagrams (in situ coordinates). (x) Equal-area plots (in situ coordinates) showing that the high-temperature component is distributed along a great circle.

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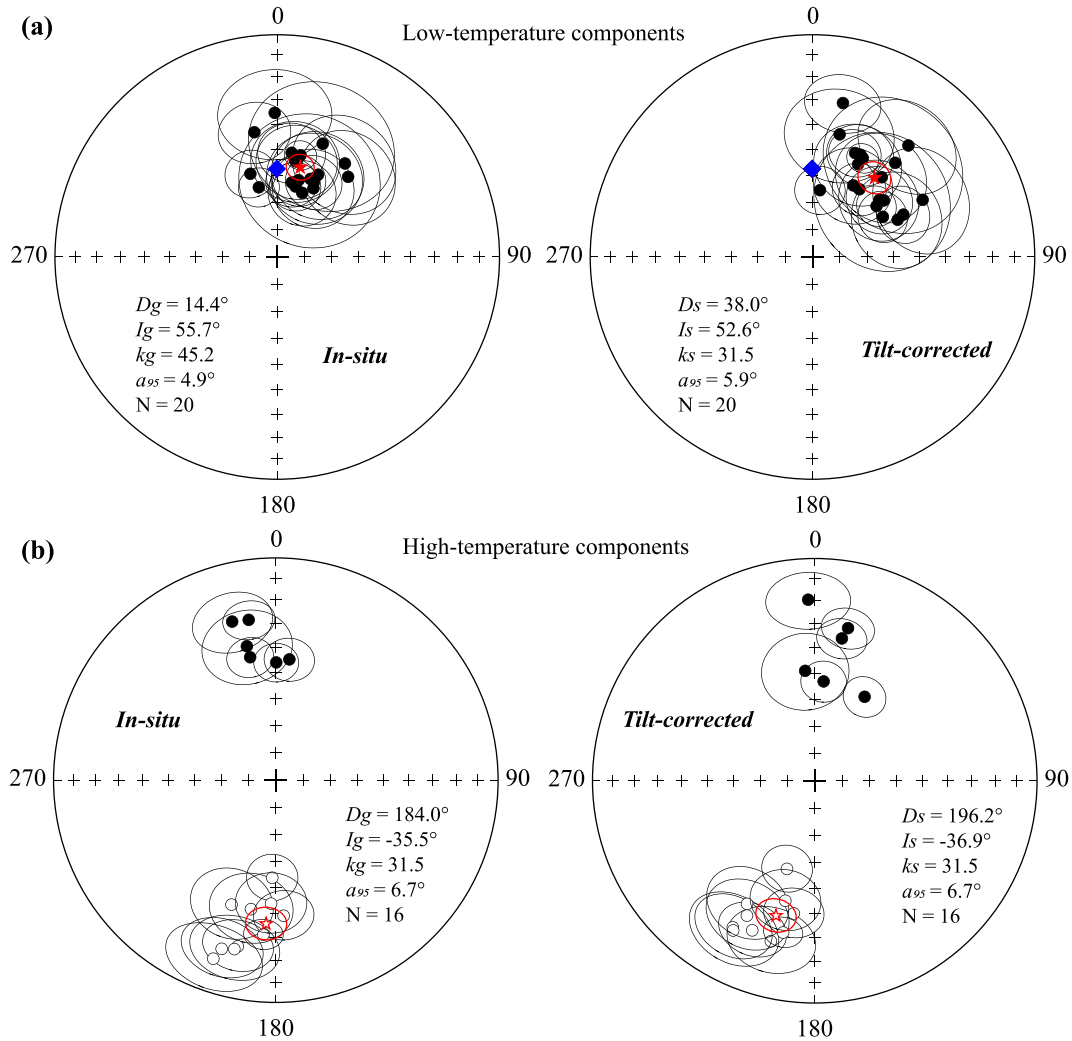


Figure 3. Equal-area projections of the direction of the magnetic components of the Late Cambrian limestone from the Oulongbuluke terrane. (a) Low-temperature component direction; (b) high-temperature component direction. Red stars indicate the mean direction. Open symbols denote upward inclinations, and solid symbols denote downward inclinations. Blue rhombus denotes the present-day geomagnetic field direction.

characteristics of carbonate rocks carrying a primary remanent magnetization. In addition, the Late Cambrian paleomagnetic poles obtained in this study are distinguished from the younger poles of the terranes in the northern Tibetan Plateau. The previous paleomagnetic poles of the terranes in the northern Tibetan Plateau are listed Table S1 in Supporting Information S1. The reliability of these paleomagnetic poles was reviewed according to the criterion for paleomagnetic data proposed by Meert et al. (2020). The results with R-score <5 are excluded from further discussion. Comparison of the dependable Ordovician to Neogene paleomagnetic poles from the Oulongbuluke, Qilian, and Qaidam terranes shows that the Late Cambrian paleomagnetic poles obtained in this study are distinguished from the younger poles (Figure S7 in Supporting Information S1).

In conclusion, the HTC obtained from the Late Cambrian limestone in the OT is a primary remanent magnetization. Dual polarities were observed in the paleomagnetic results in this study. This implies a paleolatitude of 20.6 ± 4.6°S or 20.6 ± 4.6°N for the OT during the Late Cambrian. However, paleomagnetic results indicate that the OT was located at a low latitude near the equator during the Early Ordovician (Cao et al., 2021) and at ∼27°N during the Carboniferous (Cao et al., 2017). Thus, if the OT had been located at 20.6 ± 4.6°N during the Late Cambrian, it must have drifted southward by ∼1,000 km between the Late Cambrian and the Early Ordovician, and then northward by ∼2,000 km between the Early Ordovician and the Carboniferous. Such an anomalous movement is incompatible with the surrounding continents. Furthermore, the reconstruction of the Proto-Tethys

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Table 1
Site-Mean Paleomagnetic Direction of the High-Temperature Components From the Late Cambrian Limestone in the Oulongbuluke Terrane

Sites	Strike (°)	Dip (°)	n/nt	R/N	*Dg* (°)	*Ig* (°)	*Ds* (°)	*Is* (°)	κ	α_{95} (°)	*Plat.* (°N)	*Plong.* (°E)	*dp/dm* (°)
TC12	338	31	4/6	0/4	344.6	26.9	357.9	19.6	53.0	12.7	62.8	281.0	6.9/13.3
TC13	308	16	6/6	1/5	190.9	−40.6	195.3	−26.1	23.9	14.0	−62.8	62.4	8.2/15.1
TC14	308	16	7/8	0/7	6.4	44.5	12.3	30.4	52.0	8.5	66.4	245.8	5.3/9.5
TC15	28	20	6/6	4/2	192.9	−24.3	202.7	−27.9	26.6	11.0	−59.8	48.6	6.6/12.0
TC16	32	26	5/7	0/5	0.2	46.0	30.7	53.8	102.2	7.6	65.0	184.0	7.4/10.6
TC17	32	26	8/8	4/4	197.8	−21.4	209.2	−25.4	10.8	12.7	−54.6	41.4	7.4/13.7
TC18	42	26	7/7	3/4	193.8	−22.8	206.7	−32.6	11.8	13.9	−59.5	39.1	8.9/15.7
TC19	42	26	5/8	3/2	199.3	−16.0	208.6	−24.3	13.6	14.6	−54.6	42.9	8.4/15.6
TC20	30	18	3/6	0/3	348.2	42.9	5.1	53.0	246.9	7.9	84.4	226.9	7.6/10.9
[a]TC21	30	18	6/6	0/6	343.6	46.5	2.2	57.5	7.9	25.3	88.1	160.4	27.1/37.0
TC35	20	11	10/11	1/9	182.1	−53.8	197.5	−55.8	25.0	8.9	−76.0	5.1	9.1/12.8
TC36	5	15	9/9	1/8	176.7	−38.8	189.0	−39.4	21.9	10.0	−73.1	66.6	7.2/12.0
TC37	46	13	6/8	0/6	347.8	38.3	355.0	48.9	20.9	15.0	81.4	307.0	13.1/19.8
TC40	8	12	6/8	3/3	199.6	−40.3	208.8	−36.9	23.9	14.0	−60.0	32.3	9.6/16.4
TC41	8	12	8/8	8/0	184.6	−30.9	191.8	−30.9	31.3	6.3	−66.9	66.6	3.9/7.0
TC42	8	12	4/5	1/3	182.2	−43.7	193.8	−43.7	22.9	12.0	−73.4	47.6	9.3/15.0
[a]TC43	11	17	6/7	1/5	176.6	−52.1	199.1	−53.0	9.8	18.7	−74.0	14.3	17.9/25.9
TC44	13	33	6/6	0/6	350.4	27.7	10.8	35.1	68.2	8.2	69.7	245.8	5.5/9.5
Mean direction			**100/117**	**29/71**	**184.0**	**−35.5**			**31.5**	**6.7**			
							196.2	**−36.9**	**31.5**	**6.7**			

Note. Strike, dip direction of beds minus 90°; Dip, the dip azimuth of beds; *n*, number of samples used to calculate mean direction; nt, number of samples measured; *R*, number of samples with reversed polarity; N, number of samples with normal polarity; *Dg*, declination in geographic coordinates; *Ig*, inclination in geographic coordinates; *Ds*, declination in stratigraphic coordinates; *Is*, inclination in stratigraphic coordinates; κ, the best estimate of the precision parameter; α_{95}, the radius that the mean direction lies within 95% confidence; Plat., latitude of paleomagnetic poles, positive numbers are northern hemisphere paleolatitudinal positions, negative numbers are southern hemisphere paleolatitudinal positions; Plong., longitude of paleomagnetic poles; *dp*/*dm*: semi-axes of the 95% probability ellipse around the mean pole. [a]Sites not used to calculate Fisher statistics, because α_{95} is > 15°. Reversal test: Statistics (*r*) = 11.9° < critical gamma (rc) = 13.1°, passes the reversal test with a C classification at the 95% confidence level (McFadden & McElhinny, 1990).

Ocean based on geological evidence shows that the terranes in the northern Tibetan Plateau were located in the Southern Hemisphere during the early Paleozoic (Dong et al., 2021; Li et al., 2018). Therefore, the OT should have been located at 20.6 ± 4.6°S during the Late Cambrian. The sedimentary characteristics of storm deposits within the studied area imply that the OT was located at a low paleolatitude during the Paleozoic (Ma et al., 2017). Furthermore, red beds recognized in the Late Cambrian stratum and its sedimentological characteristics indicate that the OT was also located at a low paleolatitude during the Late Cambrian (Li, Lin, et al., 2019). As mentioned above, the paleogeographic position of the OT during the Late Cambrian inferred from the new paleomagnetic results is consistent with the sedimentological evidence.

4.2. Late Cambrian Paleogeography of the Proto-Tethys Ocean

During the Late Cambrian, the main body of the Gondwana continent was assembled in the Southern Hemisphere; Laurentia was located at low equatorial latitudes; Baltica was located at ~40–70°S; and Siberia was located between 30°S and the equator (Torsvik et al., 2012). The Apparent Polar Wander Path (APWP) reported by Huang et al. (2018) suggests that the North China, South China, and Tarim blocks were located within 30° of the equator in the Southern Hemisphere during the Late Cambrian. The three blocks were located on the western margin of East Gondwana (Huang et al., 2018). The SCB may have been part of Gondwana and connected to the northwestern margin of Australia during the Late Cambrian (Li et al., 2018; Yang et al., 2004). The Lhasa, Qiangtang, and Indochina terranes were collectively located at the Gondwana margin during the Early Paleozoic (Metcalfe, 2011).

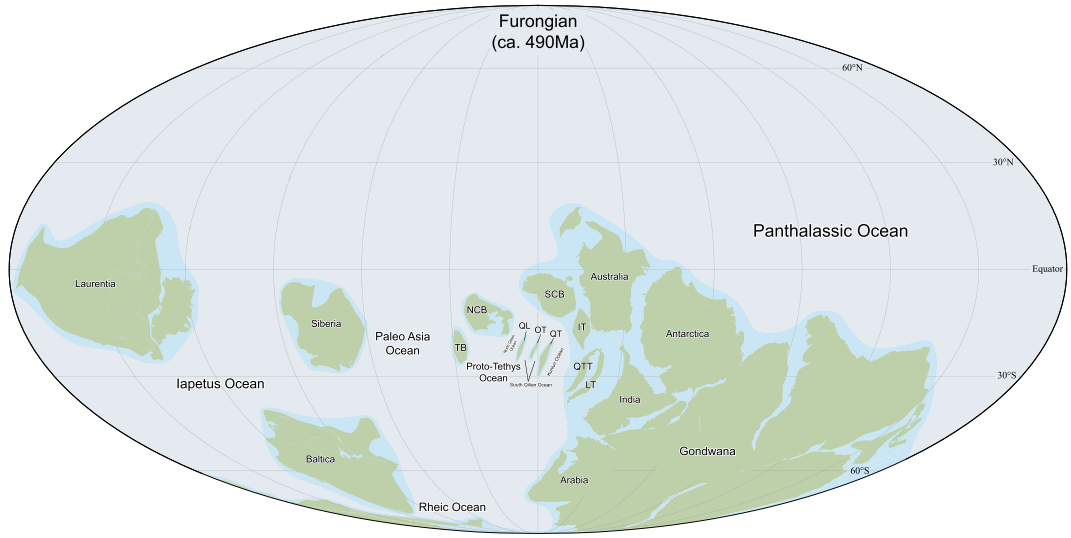


Figure 4. Paleogeographic reconstruction of the Proto-Tethys Ocean in a global setting during the Late Cambrian. NCB, North China block; SCB, South China block; TB, Tarim block; QT, Qaidam terrane; OT, Oulongbuluke terrane; QL, Qilian terrane; QTT, Qiangtang terrane; LT, Lhasa terrane; IT, Indochina terrane. Gondwana, Laurentia, Baltica, and Siberia are shown according to Torsvik et al. (2012), and North China, South China, and Tarim are shown according to Huang et al. (2018). The Oulongbuluke terrane is shown according to its Late Cambrian paleolatitude obtained in this study. The orientation and relative location of the Oulongbuluke, Qaidam, and Qilian terranes are shown according to Dong et al. (2021). Absolute longitudes of the terranes are arbitrary.

The spatial distribution of the terranes in the northern Tibetan Plateau in the Early Paleozoic was similar to their present frame (Sun et al., 2020), and the Qaidam and Qilian terranes may have been close to the OT (Song et al., 2014). During the Early Paleozoic, the OT was separated from the Qilian and Qaidam terranes by the South Qilian Ocean, a branch of the Proto-Tethys Ocean (Dong et al., 2021; Li, Wang, et al., 2019; Song et al., 2006). The South Qilian Ocean extended in a near south-north direction, and the Qaidam, Oulongbuluke, and Qilian terranes were emplaced along an east-west direction during the Early Paleozoic (Dong et al., 2021).

Several biogeographic studies show that the trilobites identified from the Oulongbuluke, Qilian, and Qaidam terranes are typical of NCB-type trilobite assemblages (Li, Wang, et al., 2019; Sun et al., 2024), indicating the close affinity of the terranes in the northern Tibetan Plateau with the NCB. The similarity of marine red beds implies that the OT and the NCB were at similar latitudes and in similar climatic zones during the Late Cambrian (Li, Lin, et al., 2019). These results indicate that the OT may have been close to the NCB during the Late Cambrian. However, the NCB was located at ~15°S (Huang et al., 2018) or ~20°N (Zhao et al., 2021) during the Late Cambrian. There is a significant difference between the paleolatitudes of the OT and the NCB during the Late Cambrian. Furthermore, the OT rapidly drifted northward with the paleolatitudes changing from ~20°S to the equator between the Late Cambrian and the Ordovician (Cao et al., 2021). The NCB was located at the paleolatitude of ~20°S during the Late Cambrian and the Ordovician and rapidly moved northward during the Ordovician and Silurian (Huang et al., 2018). The difference in paleolatitudinal changes suggests that the OT and the NCB were separated during the Early Paleozoic.

Detrital zircon data from the Cambrian sediments in the OT show a strong similarity to those of coeval sediments in the Qiangtang terrane, the Himalaya terrane, and the SCB (Sun et al., 2020). However, the Late Cambrian paleolatitude difference between the OT and the SCBs is ~10° (Figure 4), which indicates that the OT was not affiliated with the SCB. Additionally, detrital zircon ages of the Cambro-Ordovician sandstones of the South Qilian and North Qilian tectonic belts also suggest the affinity of the Qilian terrane with the Indian plate (Li, Wang, et al., 2019; Zhang et al., 2015). Zircon geochronologic data are consistent with the paleolatitude of the OT and the Indian continent. The paleomagnetic data in this study indicate that the Late Cambrian paleolatitude of the OT was 20.6 ± 4.6°S. The Late Cambrian paleolatitude for northwestern India was ~30°S, as inferred from the Gondwana reconstruction (Torsvik et al., 2012). These results indicate that the OT was located close to northwestern India during the Cambrian.

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In summary, our results enable a tentative paleogeographic reconstruction of the OT during the Late Cambrian (Figure 4). This reconstruction suggests that the terranes in the northern Tibetan Plateau may have been located in northwestern India of Gondwana during the Late Cambrian. The OT was located between the Qaidam and the Qilian terranes, and these three terranes were located close to northwestern India and distributed in an east-west direction. The Qaidam and Qilian terranes were separated by the South Qilian Ocean and the OT was located within this ocean during the Late Cambrian. The Qilian terrane was separated from the Tarim and North China blocks by the North Qilian Ocean. The South Qilian Ocean and the North Qilian Ocean were both branches of the Proto-Tethys Ocean.

5. Conclusions

We have obtained the first Late Cambrian paleomagnetic results for the OT. The mean paleomagnetic direction in stratigraphic coordinates is $Ds = 196.2°$, $Is = -36.9°$ with $\kappa s = 31.5$, $\alpha_{95} = 6.7°$, $N = 16$, corresponding to a paleopole at 68.2°S/51.9°E with dp/dm = 4.6°/7.8°. Rock magnetic and microstructural analyses indicate the primary origin of the magnetic minerals in the studied section. Combining the paleomagnetic results with geologic evidence from the northern Tibetan Plateau suggests the following: (a) The Late Cambrian paleolatitude of the OT was 20.6 ± 4.6°S (reference point: 37.2°N/96.6°E); and (b) the terranes in the northern Tibetan Plateau may have been located to the northwest of the Indian plate of Gondwana during the Late Cambrian.

Data Availability Statement

Data to support this study are available in Supporting Information S1 and Cao et al. (2024).

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